MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY CHAPTER TEST – 04** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Marks are indicated against each question.
2. **Section A** consists of 5 questions of 1 mark each (4 MCQs and 1 Assertion-Reason).
3. **Section B** consists of 4 questions of 11 marks total (1 VSAQ of 2 marks and 3 SA of 3 marks each).
4. **Section C** consists of 1 Case-Based integrated unit of assessment of 4 marks with sub-parts.
5. Use of calculators is strictly prohibited. Neat geometric sketches must accompany geometric proofs.

SECTION A: MULTIPLE CHOICE QUESTIONS (1 Mark Each)

1. In trapezium $ABCD$, $AB \parallel DC$ and diagonals intersect at O. If $AO = (3x - 1)$, $OC = (5x - 3)$, $BO = (2x + 1)$, and $OD = (6x - 5)$, the value of x is:

(a) $\frac{1}{2}$ (b) 2 (c) 3 (d) 4

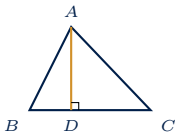
2. In ΔABC, D and E are points on sides AB and AC such that $AD = 2$ cm, $BD = 3$ cm, $AE = 3.6$ cm, and $AC = 9$ cm. Then:

(a) DE is not parallel to BC (b) $DE \parallel BC$ (c) $DE \perp BC$ (d) $\angle ADE = \angle B$ is false

3. If $\Delta ABC \sim \Delta QRP$, $\frac{AB}{QR} = \frac{1}{2}$, $BC = 2.5$ cm, and $CA = 3$ cm ($AB = 3$ cm), the perimeter of ΔPQR is:

(a) 15 cm (b) 17 cm (c) 16 cm (d) 20 cm

4. In ΔABC, $\angle A = 90°$ and $AD \perp BC$. If $BD = 3.6$ cm and $CD = 6.4$ cm, then the length of altitude AD is:



(a) 4.8 cm (b) 5.0 cm (c) 4.2 cm (d) 5.4 cm

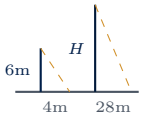
5. Assertion (A): In ΔABC, D and E are on AB, AC such that $AD = 3, DB = 4.5, AE = 2, EC = 3$. Then $DE \parallel BC$.
Reason (R): If a line divides two sides of a triangle in the same ratio, it is parallel to the third side.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

SECTION B: DESCRIPTIVE QUESTIONS (PART 1)

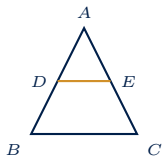
6. **[2 Marks]** E and F are points on sides PQ and PR respectively of ΔPQR. If $PE = 3.9$ cm, $EQ = 3.0$ cm, $PF = 3.6$ cm, and $FR = 2.4$ cm, determine whether $EF \parallel QR$, stating the mathematical criterion used.

7. **[3 Marks]** A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.

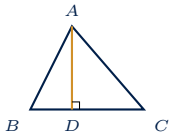


SECTION B: DESCRIPTIVE QUESTIONS (PART 2 Continued)

8. **[3 Marks]** In the given figure, $\angle D = \angle E$ and $\frac{AD}{DB} = \frac{AE}{EC}$. Prove mathematically that ΔBAC is an isosceles triangle.

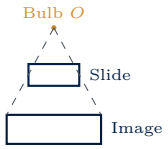


9. **[3 Marks]** In ΔABC, $AD \perp BC$ such that $AD^2 = BD \times CD$. Prove mathematically that ΔABC is a right-angled triangle at A.



SECTION C: CASE STUDY BASED QUESTION (4 Marks)

10. **Case Study 04: Overhead Optical Projector & Quadrilateral Enlargement**



An overhead projector bulb is fixed at point O above a presentation table. A transparent slide in the form of quadrilateral $ABCD$ is held horizontally between the bulb and the table top. The projected image forms an enlarged quadrilateral $A'B'C'D'$ on the screen with a scale factor of $1 : 2.5$. The dimensions of the slide are $AB = 2.0$ cm, $BC = 3.0$ cm, $CD = 2.4$ cm, $DA = 1.6$ cm, and $\angle A = 105°$.

(i) **[1 Mark]** What is the exact measure of $\angle A'$ in the projected image?

(ii) **[1 Mark]** Find the length of side $B'C'$ of the projected image.

(iii) **[2 Marks]** Calculate the total perimeter of the projected quadrilateral $A'B'C'D'$.

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